The Perks of Being a First Mover

Does White's Advantage in Chess Grow or Shrink with Rating?

CS215 Open Project · Checkpoint 1

1. Meta Information

Project title	The Perks of Being a First Mover
Group number	02
Mentor	Namish Bansal
Members	Sanidhya (25b0944), Daksh Maahor (25b0974), Sahil (25b0991), VadaPavJai (25b1030)
Repository	https://github.com/Sahil-M-Deo/The-Perks-of-Being-a-First-Mover
Tracking sheet	`https://docs.google.com/spreadsheets/d/1pvTllHR4a1Sl7GHtjKnvLeb-zmmHaM3BjltyaDPX9-Q`

2. Progress So Far

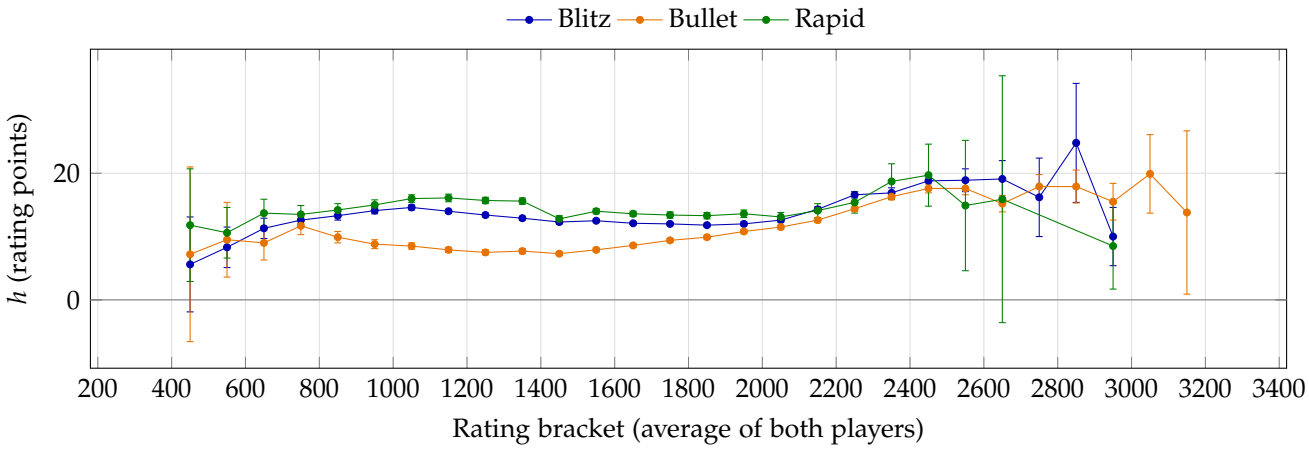
Dataset and descriptive statistics

Source: Lichess Open Database (`https://database.lichess.org`), standard rated games for 2024-01 and 2024-02, released under the Creative Commons CC0 licence, so it is free to download, use and redistribute. Each month is one zstd-compressed PGN file of tens of GB (well over 100 GB once decompressed). Every game is one record with about 17 header tags (players, ratings, result, date, opening, time control, termination) followed by the move list. We use 6 tags: `Result`, `WhiteElo`, `BlackElo`, `Event` (gives the speed), `TimeControl` and `Termination`.

Month	Games read	Games kept	Kept
2024-01	98,994,760	84,177,790	85.0%
2024-02	91,567,975	77,997,622	85.2%
Total	190,562,735	162,175,412	85.1%

Speed	Games	White score	Expected	h (pts, 95% CI)
Blitz	77,918,475	0.518	0.500	$+12.8 \pm 0.1$
Bullet	59,945,392	0.514	0.500	$+9.9 \pm 0.1$
Rapid	22,650,621	0.520	0.500	$+14.2 \pm 0.1$
Classical	857,174	0.519	0.500	$+13.4 \pm 0.7$
UltraBullet	722,976	0.513	0.500	$+9.1 \pm 0.8$
Correspondence	80,774	0.504	0.500	$+2.8 \pm 2.4$

White score is White's average score (win 1, draw $\frac{1}{2}$, loss 0); *Expected* is the score the two ratings alone predict. Over all kept games White scores 0.517 against an expected 0.500, which corresponds to $h = +11.9 \pm 0.1$ rating points. Within Blitz, h ranges from +5.6 in bracket [400,500) to +24.8 in [2800,2900) across 26 brackets (figure below; brackets with fewer than 1,000 games are hidden).



Data cleaning steps completed

- **Parser:** a C++ program (`scripts/white_advantage.cpp`) streams each compressed file through the zstd library in 1 MB chunks, reads only header lines and skips the moves. Memory stays at a few MB, and the two months are processed in parallel, one thread each.
- **Removed:** games with no result (6,007), abandoned games (502,851), games with a missing or "?" rating (0), and games where the rating gap is over 100 points (27,878,465).
- **Grouped:** by speed (UltraBullet to Correspondence, from the `Event` tag) and by 100-point rating bracket $[a, a+100)$ of the *average* of both ratings, so every game falls in exactly one bracket.
- **Aggregated:** instead of exporting one CSV row per game, we store game counts and White's total score for every (speed, bracket, rating gap) cell. This is all the model needs, and it fits in a few MB.
- **Checked:** on synthetic games with a known built-in advantage of $+35$ points, the pipeline recovers $+34.8 \pm 1.7$.

Division and revision of responsibilities

Member	Responsibility (revised since the proposal: work split equally)
Sanidhya	Statistical modelling: fitting h and its confidence intervals; next, the trend test across brackets.
Daksh Maahor	Data cleaning: filters, speed and rating-bracket grouping; next, separate win/draw rates per bracket.
Sahil	Data engineering: C++ streaming parser for the `.pgn.zst` files; next, robustness runs across time controls.
VadaPavJai	Visualisation and report: figures and this report; next, the Plotly dashboard.

Refined problem statement

The proposal compared raw win rates across Elo brackets. Raw win rates are confounded by the rating gap: a 1540 who beats a 1570 with White is stronger evidence of a first-move edge than a win between two 1570s. We therefore model each game with the Elo expected-score curve plus a White bonus h:

$$E = \frac{1}{1 + 10^{-(R_{\text{White}} + h - R_{\text{Black}})/400}},$$

and estimate h, White's advantage in rating points, by maximum likelihood separately for each speed and each 100-point bracket, with a 95% confidence interval from the likelihood curvature. Draws count as half a point. The question becomes: *does h rise to a peak at intermediate ratings and fall at 2200+?* We also widened the rating-gap filter from 50 to 100 points, since the model now corrects for the gap itself and the wider filter keeps far more games.

For Checkpoint 2: a formal test for a trend in h across brackets, separate win and draw rates per bracket (draws rise with rating), robustness across time controls, and a Plotly dashboard.

Tracking spreadsheet

Updated with Checkpoint 1 rows (meta information, dataset, cleaning, revised roles, refined problem statement); the mentor has edit access. Link in Section 1.

Blockers and challenges

- **Raw data can't go on GitHub.** The guidelines ask us to push data files, but each monthly file is tens of GB, far above GitHub's limits (100 MB per file, 2 GB with Git LFS on the free plan). We are pushing the aggregated output (`output/summary.txt`, `output/output.csv`) and the exact download links instead. Is this acceptable?
- The two concerns from our proposal are resolved: streaming the files avoids the memory limit, and draws are handled as half a point inside the model.